MONTE CARLO METHODS IN EXPERIMENTAL PHYSICS

# Monte Carlo simulation of $\pi^-$ interactions on a polyethylene target and optimization of the target thickness

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# Introduction

- Monte Carlo simulations are a useful tool in experimental high-energy physics, where detector design and analysis strategies must be optimized before data taking.
- By generating synthetic events according to known physical laws and detector responses, Monte Carlo methods allow to study the experimental reconstruction and its statistical performance in a controlled environment
- In this work, a Monte Carlo simulation of a $\pi^-$ beam with an energy of 100 GeV impinging on a polyethylene target is developed.
- The signal processes $\pi^- p \to \pi^0 n$ and $\pi - p \to \eta n$, followed by the decay $\pi^0 \to \gamma\gamma$ and $\eta \to \gamma\gamma$, are considered.
- The main goal of the study is the **optimization of the target thickness** $L$, in order to **maximize the statistical significance** of the reconstructed signal while preserving a good **invariant-mass resolution**.

# Experimental setup

- **Incident beam** → 100Gev nominal, no beam divergence, single relevant interaction per incident pion (neglects secondary hadronic interactions inside the target), no pile-up effects
- **Target** → $CH_2$ , homogeneous medium, only hydrogen protons at rest considered for the interaction
- **Signal vs background** → only $\pi^0/\eta \to \gamma\gamma$ process are signal contributions. All other interactions are treated as background
- **Detector** → sensitive exclusively to $\gamma$ final states. No information on the recoiling neutron or on the primary interaction vertex. The optimization procedure is entirely based on the reconstruction of the invariant mass $m_{\gamma\gamma}$
- **Target and detector geometry**
  **Target** → cylinder of $R$ = 10 cm, length $L$ , axis aligned along beam direction
  **Detector** → circular surface of $R$ = 100 cm, centered on beam axis, located at $z$ = 10m from target

# Interaction depth sampling

- **Probability density** for the first interaction to occur at depth $z$ $\longrightarrow$ $P(z)=\frac{1}{\lambda}\exp\left(-\frac{z}{\lambda}\right)$
- **Interaction length**: $\lambda=\frac{1}{n\sigma_{tot}}\approx 499.95\,cm$ where $\longrightarrow$ $n_p=2\frac{\rho}{M}N_A\approx 8.00\cdot 10^{22}\,cm^{-3}$

  $\longrightarrow$ $\sigma_{tot}\,(\pi^- p, 100GeV)\approx 25\,mb$
- The **probability** for a pion to undergo **at least one interaction** within a target of thickness $L$ :

$\longrightarrow$ $P_{inter}(L)=\int_0^L \frac{1}{\lambda}e^{-x/\lambda}=1-e^{-L/\lambda}$ ; If $L\ll\lambda$ $\longrightarrow$ $P_{inter}(L)\approx\frac{L}{\lambda}$

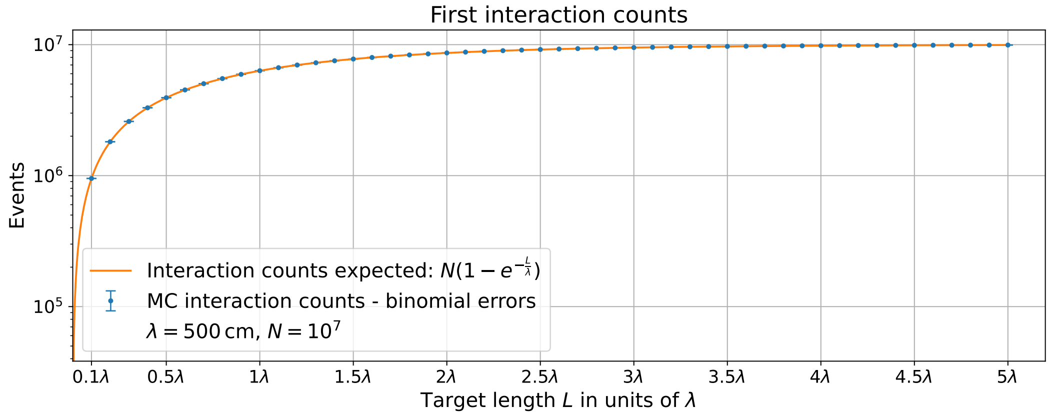
- Approximation: $z$ is therefore used as the **interaction position** along the target: $L_{meson}\simeq\frac{E}{m}c\tau\ll\lambda$
- $z$ **sampling** from **inverse transform method**:

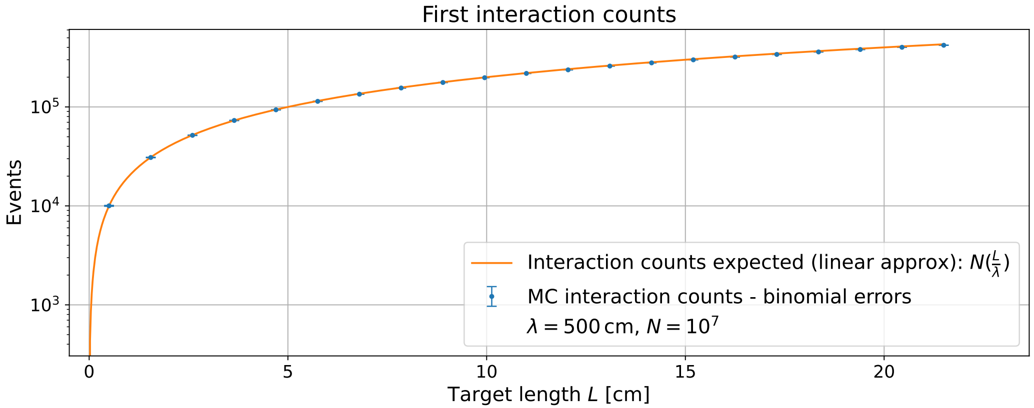
$\longrightarrow$ the CDF $F(z)=P(Z\leq z)$ is uniformly distributed in [0,1]

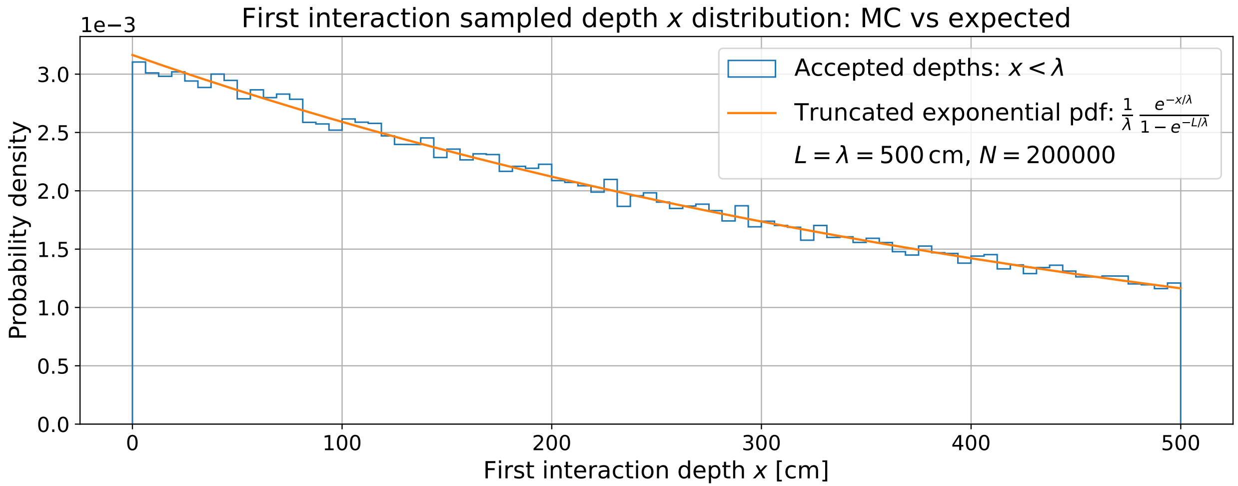
$\longrightarrow$ $u=F(z)=1-e^{-z/\lambda}$ $\longrightarrow$ $\boxed{z=-\lambda\ln u}$

# Interaction depth sampling validation

Monte Carlo vs analytical prediction for the **number of first interactions** vs target thickness $L$ .



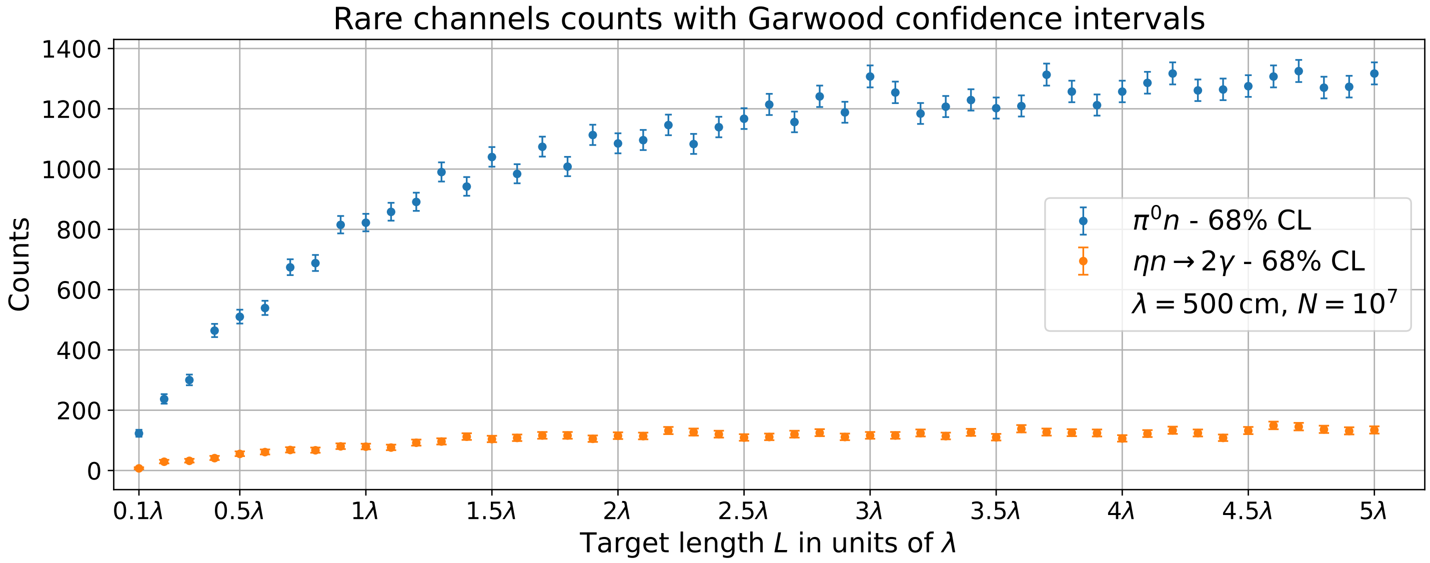




Distribution of the **sampled interaction depth $z$** for a fixed target thickness, compared to the corresponding **truncated exponential probability density**.
Small deviations are due to finite Monte Carlo statistics and binning effects.

# Channel probabilities

- Given an interaction inside the target, the final state of the event is selected probabilistically according to the **relative production cross sections** of the available channels

$$p_\eta = \frac{\sigma_{2\gamma}(\pi^- p \to \eta n)}{\sigma_{tot}(\pi^- p, 100\,GeV)} \approx 0.001\,\% \quad ; \quad p_{\pi^0} = \frac{\sigma_{tot}(\pi^- p \to \pi^0 n)}{\sigma_{tot}(\pi^- p, 100\,GeV)} \approx 0.013\,\% \quad ; \quad p_{bkg} = 1 - p_\eta - p_{\pi^0} = 99.9\,\%$$




Expected event counts for the rare $\pi^- p \to \pi^0 n$ and $\pi^- p \to \eta n$ channels vs target thickness $L$. Error bars represent 68% confidence intervals computed using the **Garwood construction** for Poisson distributed counts.

# Meson angular distribution

- In the process π⁻ p → X n the information on the angular distribution in the center-of-mass frame is encoded in the **differential cross section** $d\sigma/d\tau$ with $\tau \equiv -t$
- From tabulated values of $d\sigma/d\tau$ → $\Delta\sigma_i \simeq y_i \Delta\tau_i$ with $\begin{cases} y_i \equiv \left(\frac{d\sigma_i}{d\tau_i}\right)_i \\ \Delta\tau_i \end{cases}$ → Differential cross-section **constant** within each bin; → Bin width
  $\Delta\sigma_i$ corresponding to the bin area in the $(\tau, d\sigma/d\tau)$ plane
- The **probability** for an event **to fall into bin $i$** is defined as the fraction of the total cross section carried by that bin

$$\longrightarrow P_i = \frac{\Delta\sigma_i}{\sum_{j=1}^{N} \Delta\sigma_j} \quad \text{with} \quad \sum_{i=1}^{N} P_i = 1$$

- The resulting **probability density function for $\tau$** is piecewise constant

$$p(\tau) = \begin{cases} \dfrac{P_i}{\Delta\tau_i} = \dfrac{y_i}{\sum_j y_j \,\Delta\tau_j} & \tau \in [\tau_i^{\text{low}}, \tau_i^{\text{high}}], \\ 0 & \text{otherwise} \end{cases}$$

# Meson angular distribution

- **Discrete CDF** from the bin probabilities → $CDF_k = \sum_{i=1}^{k} P_i \quad ; \quad CDF_N = 1$

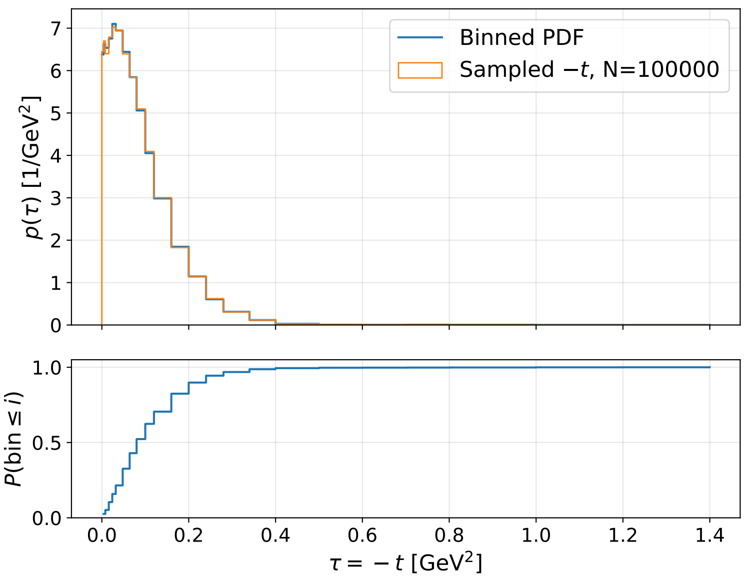
  The value **CDF$_k$** represents the **probability** that an event **belongs to a bin with index $\leq k$**

- **$\tau \equiv -t$ sampling**:

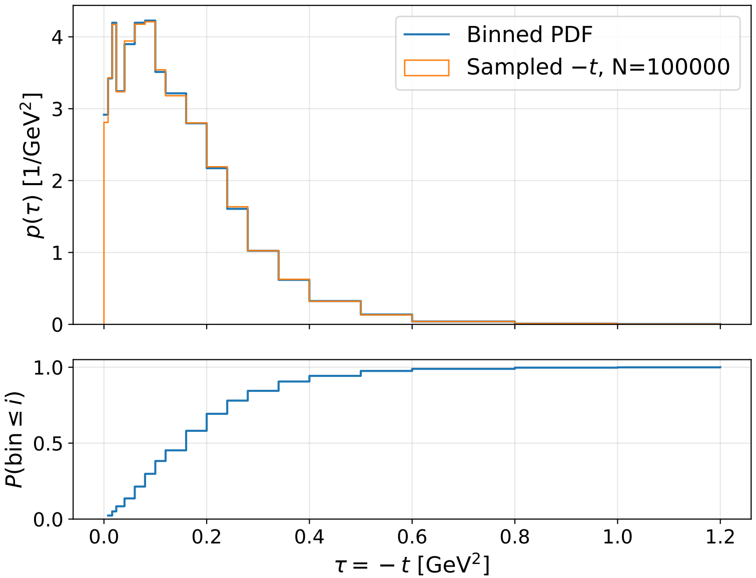
  → $u \sim \mathcal{U}[0,1]$ ; the first bin satisfying $CDF_i > u$ is selected. This assigns bin $i$ with $P_i$

  → $\tau \sim \mathcal{U}\,[\Delta\tau_{\mathrm{i}}]$

Binned pdf and CDF for t-sampling from $\frac{d\sigma_{\pi^0}}{dt}$




Binned pdf and CDF for t-sampling from $\frac{d\sigma_{\eta}}{dt}$




Piecewise-constant pdf and CDF reconstructed from the tabulated $d\sigma/d\tau$, together with Monte Carlo generated samples.
The strong concentration of events at **small values of $\tau$** exhibits the **forward nature** of the meson production at high energy

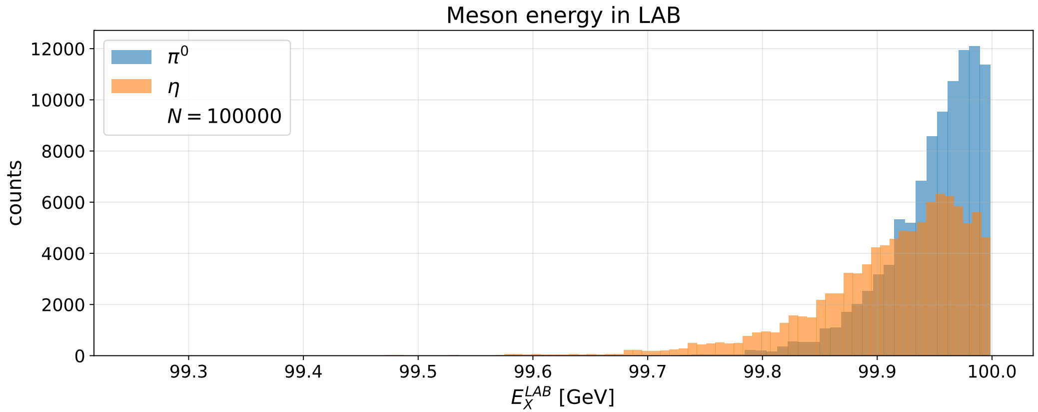
# Event kinematics

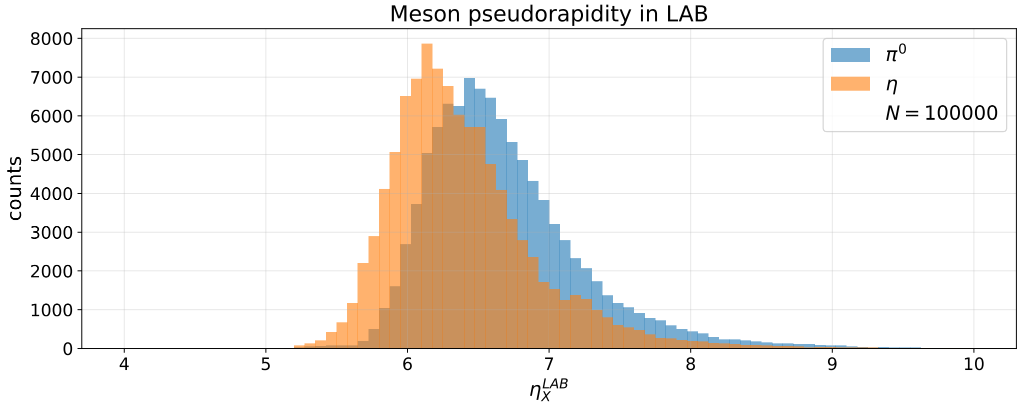
**Conventions**:

- **Polar angle $\theta$** → respect to the $z$ axis: the direction of the incoming beam
- **Azimuthal angle $\phi$** → respect to the transverse $(x,y)$ plane around the beam axis
- **Pseudorapidity $\eta$** → $\eta = -\ln\tan(\theta/2)$

**Two body scattering: $\pi^- p \rightarrow X n$**

- **Lab frame**: $s = (P_p + P_\pi)^2 = m_p^2 + m_\pi^2 + 2 m_p E_\pi$ ; $\beta_{CM} = \frac{|\overrightarrow{P_{tot}}|}{E_{tot}} = \frac{p_\pi}{m_p + E_\pi}$
- **CM frame**: $p^{CM} = \frac{\sqrt{\lambda(s, m_X^2, m_n^2)}}{2\sqrt{s}}$ ; $\cos\theta^{CM} = \frac{t - m_\pi^2 - m_X^2 + 2 E_\pi^{CM} E_X^{CM}}{2 p_\pi^{CM} p^{CM}}$ ; $\phi^{CM} \sim U[0, 2\pi]$
- Meson momentum in CM and transformation in LAB → $\vec{p}_X = p^{CM}(\sin\theta^{CM}\cos\phi^{CM}, \sin\theta^{CM}\sin\phi^{CM}, \cos\theta^{CM})$ → $p_X^{LAB} = \Lambda(\overrightarrow{\beta^{CM}})\, p^{CM}$



# Decay $X \to \gamma\gamma$

- **Meson rest frame**:

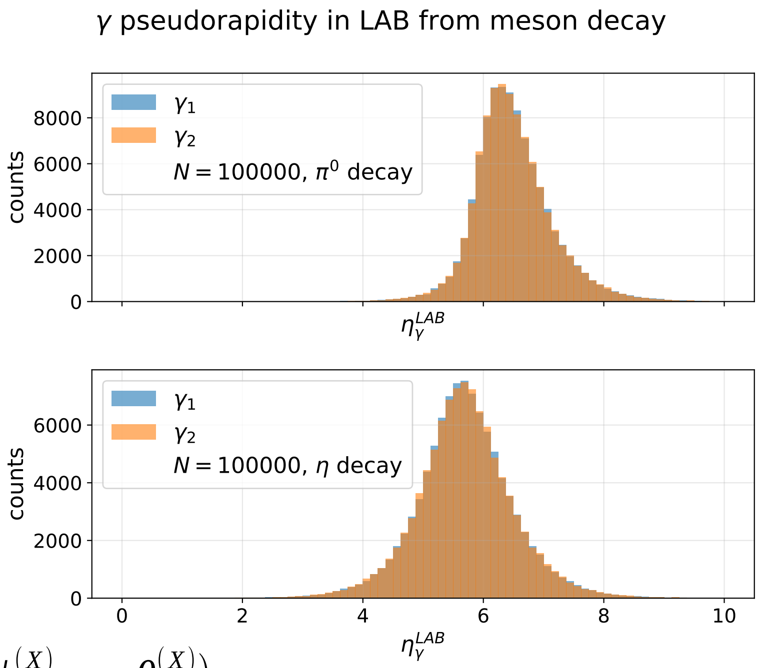
$$E_\gamma^{(X)} = \frac{m_X}{2}$$

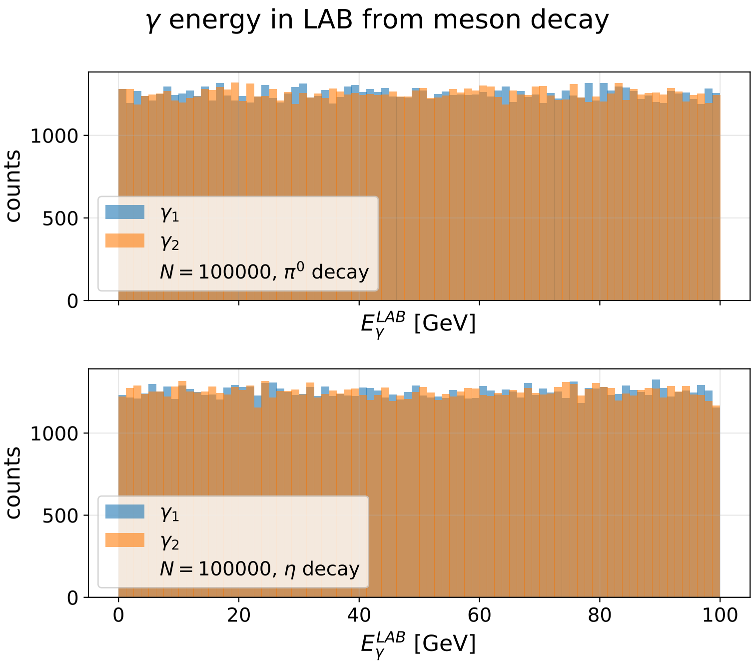
$$\cos\theta_\gamma^{(X)} \sim U[-1,1]$$

$$\phi_\gamma^{(X)} \sim U[0,2\pi]$$

$$k_{1,2}^{(X)} = \left(\frac{m_X}{2}, \pm\frac{m_X}{2}\cdot\hat{n}\right)$$

$$\hat{n} = \left(\sin\theta_\gamma^{(X)}\cos\phi_\gamma^{(X)}, \sin\theta_\gamma^{(X)}\sin\phi_\gamma^{(X)}, \cos\theta_\gamma^{(X)}\right)$$

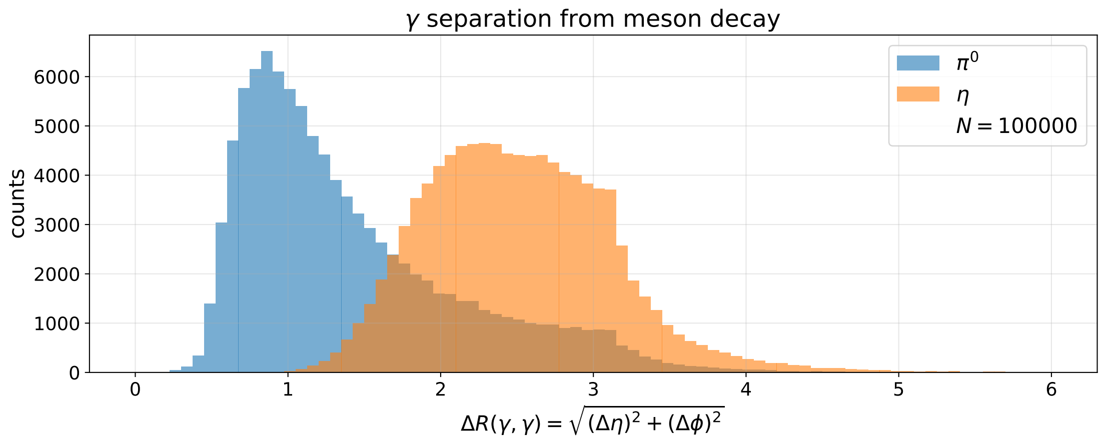



- **LAB frame**:

$$\vec{\beta}_X = \frac{\overrightarrow{p_X^{LAB}}}{E_X^{LAB}} \longrightarrow E_\gamma^{LAB} = \gamma_X E_\gamma^{(X)}\left(1 + \vec{\beta}_X\cdot\hat{n}\right)$$

- **Opening angle** in LAB between the two photons $\longrightarrow \Delta R(\gamma,\gamma) = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$



# Detector response

- **Geometry acceptance:** → a photon is accepted if its trajectory intersects the detector surface → $A_{geom}(\gamma)=\Theta\left(R_{cal}^2-x_\gamma^2-y_\gamma^2\right)$

- **Photon escape probability:** $P_{escape}(l)=\exp\left(-\frac{l}{\lambda_\gamma}\right)$ with $\lambda_\gamma$ approximated in the **electron-positron pair** production regime: → $\lambda_\gamma=\frac{9}{7}X_0\ ;\ X_0=50.31\,cm$

- **Energy smearing** → $E_\gamma^{meas}\sim N\left(E_\gamma^{true},\sigma_E^2\right)$

  worsening of the resolution induced by photon path length inside the target

  → $\frac{\sigma_E}{E_\gamma^{true}}=\sqrt{\left(\frac{a}{\sqrt{E_\gamma^{true}}}\right)^2+b_{eff}^2}$ → $b_{eff}=\sqrt{b^2+\left(k_E\frac{l}{\lambda_\gamma}\right)^2}$

- **Position smearing** → $x_\gamma^{meas}\sim N\left(x_\gamma^{true},\sigma_x^2\right)$

  worsening of the resolution induced by photon path length inside the target → $\sigma_x=\sigma_{x,0}\left(1+k\frac{l}{\lambda_\gamma}\right)$

- **Energy and angular separation threshold**

  → $E_\gamma^{meas}>E_{threshold}=0.1\,GeV$ → $\Delta R^{meas}>\Delta R_{min}=0.02$

# Event reconstruction

- **Invariant mass reconstruction** ⟶ $m_{\gamma\gamma}^2 = 2E_1^{meas} 2E_2^{meas}(1-\cos\alpha^{meas})$

  with $\cos\alpha^{meas} = \hat{n}_1 \hat{n}_2$ opening angle between the two photons

- **Background sampling** ⟶ **log-uniform distribution**, sampled $E_\gamma$ with **inverse transform method**

  ⟶ $\log E_\gamma \sim U(\log E_{min}, \log E_{max})$ ; $p(E_\gamma) = \frac{1}{E_\gamma \ln(E_{max}/E_{min})}$ ⟶ $E_\gamma = E_{min}\left(\frac{E_{max}}{E_{min}}\right)^u$ ; $u \sim U[0,1]$

- **Iterative fit on signal** ⟶ fit exclusively the signal shape with a **Gaussian model** getting $\mu_{fit}, \sigma_{fit}$

  ⟶ iterative fit in the range $\mu_{fit} \pm k\sigma_{fit}$ to get **mass resolution**

- **Oversampling** ⟶ extremely small expected number of produced neutral mesons

  ⟶ **fixed** number of generated events for signal and background

  ⟶ event reconstruction in oversampled sets is dependent from photon escape probability and detector response only, not from channel probabilities

  ⟶ To recover physical expected yields **rescale with weights**: $w_{phys} = \frac{N_{expected}}{N_{generated}}$

# Event reconstruction

- **Signal and background yields** $\longrightarrow$ $S(L), B(L)$

  number of the events in the invariant-mass oversampled spectra in an adaptive window centered at the fitted peak position $\longrightarrow$ $|m_{\gamma\gamma} - \mu(L)| < k\,\sigma(L)$

- **Uncertainty estimation**

  $\longrightarrow$ Poisson bootstrap technique: create spectra bootstrap replicas by fluctuating the content of each bin according to a Poisson distribution $n_i^{new} \sim Poiss(n_i)$ .

  $\longrightarrow$ For each bootstrap replica, the full analysis chain is repeated, getting $S(L), B(L)$ and mass resolution

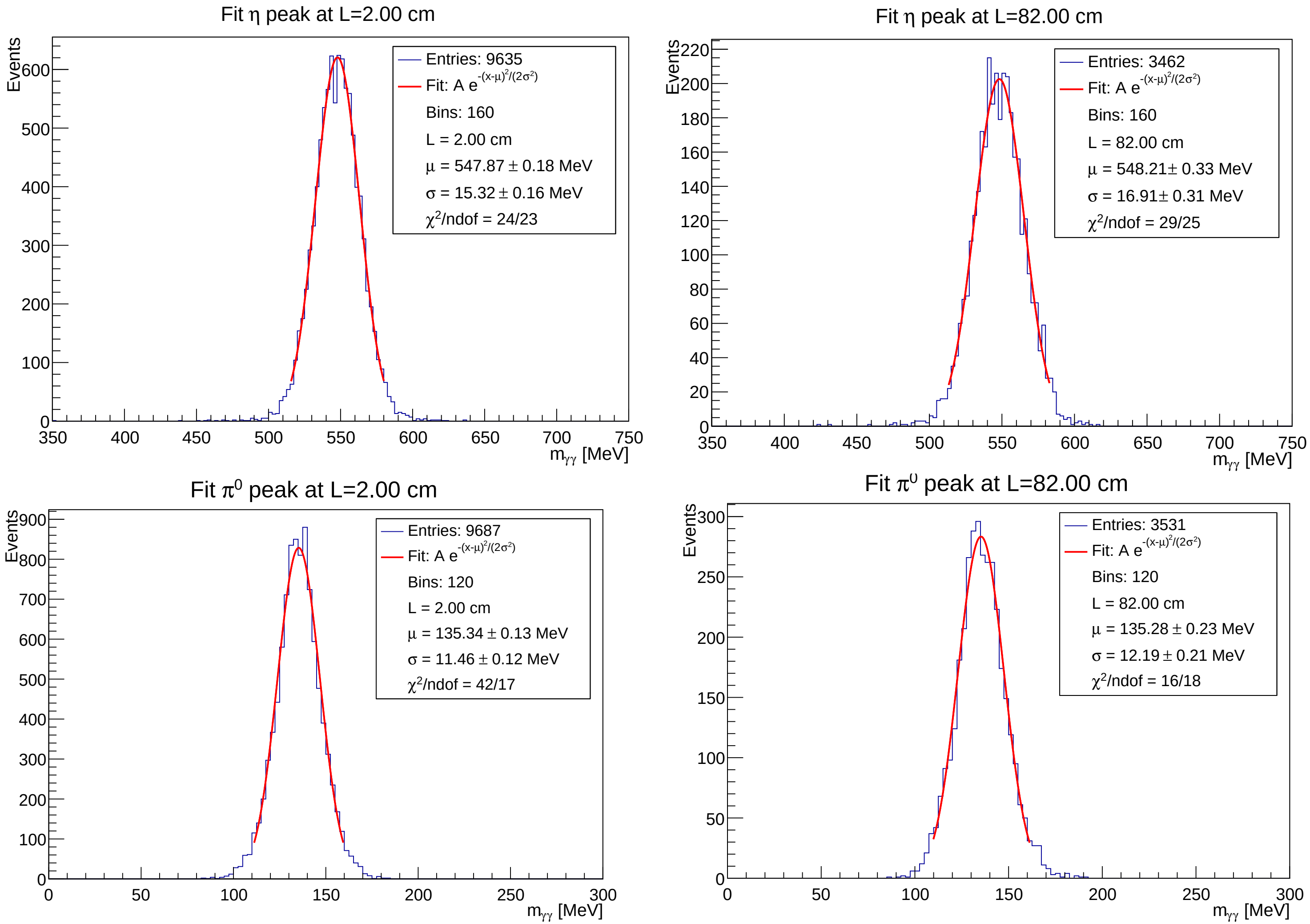
  $\longrightarrow$ Statistical uncertainty on each observable: standard deviation of its bootstrap distribution

- **Significance**

$$Z(L) = \frac{S(L)}{\sqrt{(S(L)+B(L))}}$$

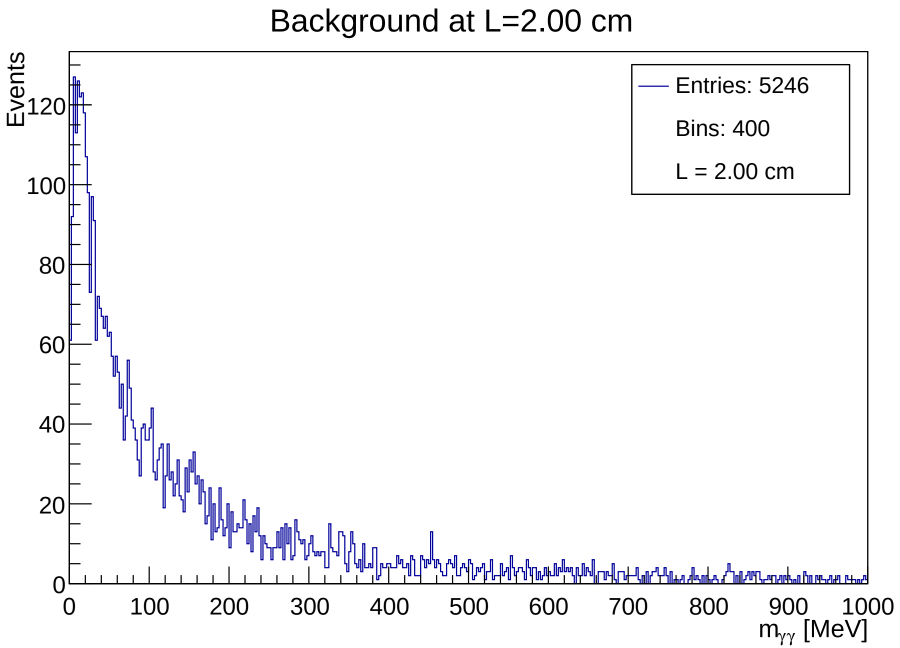
Measures how large the signal yield is compared to the typical statistical fluctuations of the total event count: estimation of the statistical signal detectability from the background

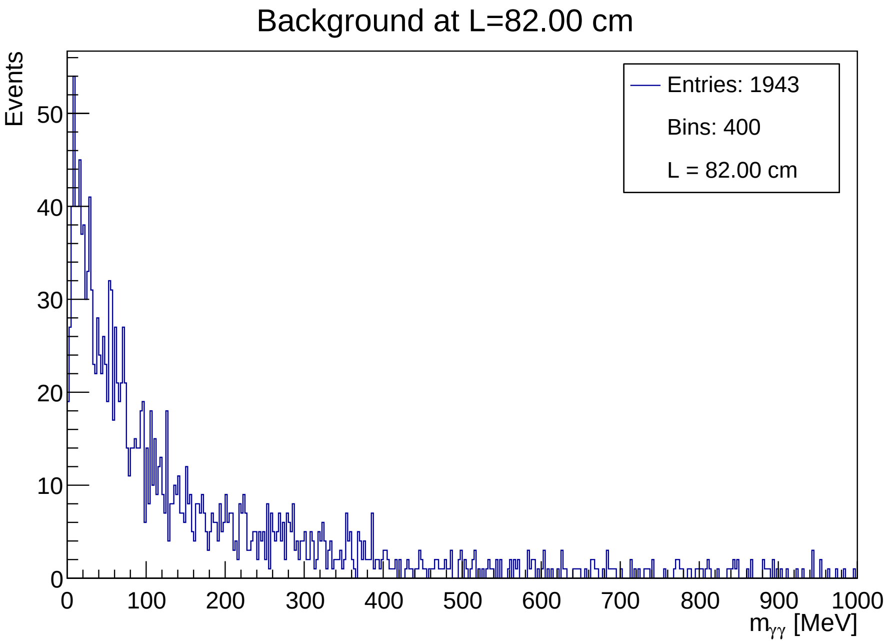
# Optimization of the target thickness




- Reconstructed **oversampled invariant-mass spectrum** for the two mesons at two representative target thicknesses.
- Total generated events: **N=10_000**
- The total number of reconstructed signal events **decreases** with increasing $L$
- The invariant-mass resolution **increases** with increasing $L$

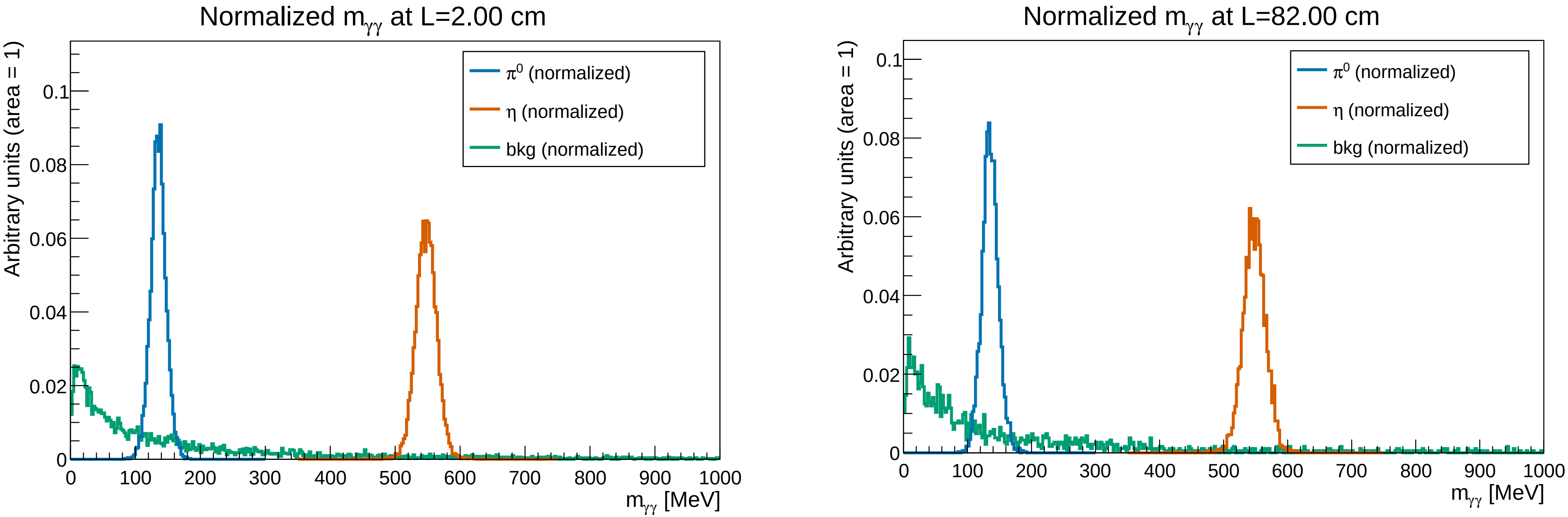
# Optimization of the target thickness



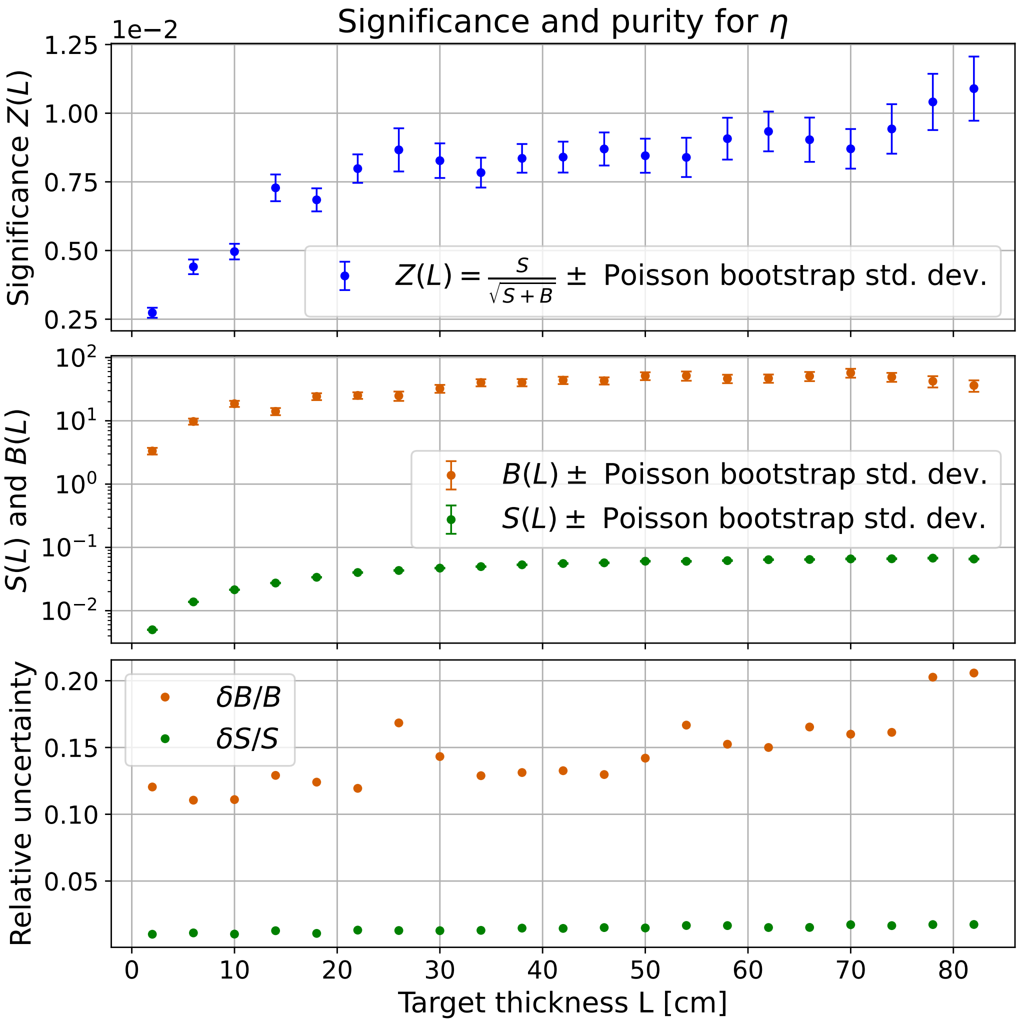

- Reconstructed **oversampled spectrum** of the **background** for two representative target thicknesses
- Total generated events: **N=10_000**
- **No resonant structure**
- **Soft photons** are produced with much **higher probability** than hard ones
- The total number of reconstructed signal events **decreases** with increasing $L$

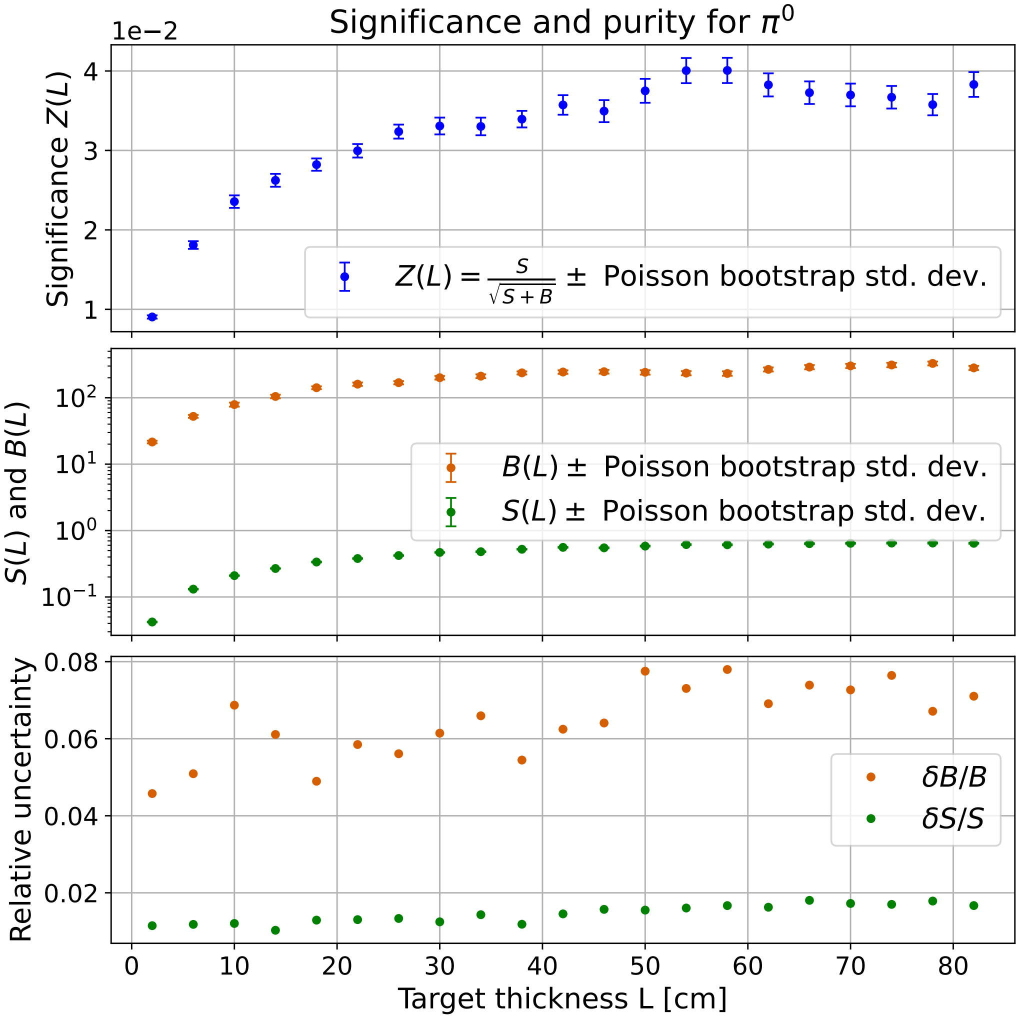
# Optimization of the target thickness



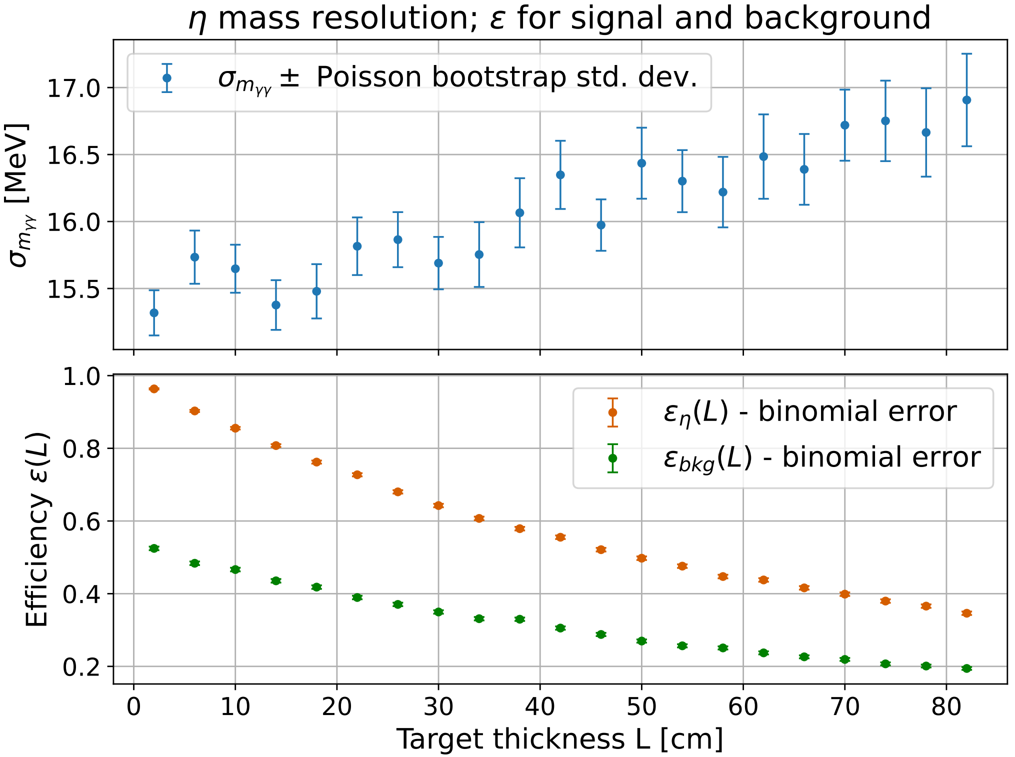

- **Normalized invariant-mass spectra** for the mesons and background
- Direct **comparison** of the **spectral shapes**
- Signals appear as **well-defined peaks** on top of a **smooth background distribution**
- **Increasing** $L$ affects the spectra in a largely proportional way: the qualitative appearance of the normalized spectra remains similar and the difference in signal shapes **is not appreciable**

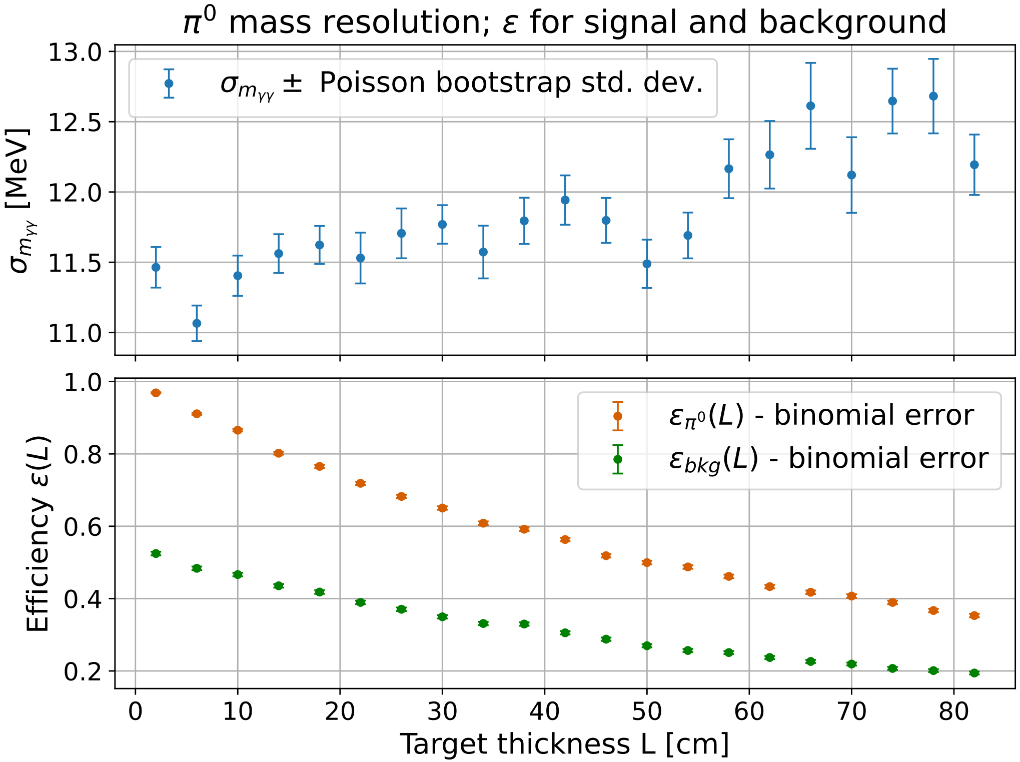
# Optimization of the target thickness



# Optimization of the target thickness




## Detector efficiency:

$$\epsilon = \frac{N_{reconstructed}}{N_{generated}}$$